Postulates & Diagrams — Practice — Answer Key

A worked example comes before each type. Show your thinking.

Name: _____ Date: _____

Type 1 Which Postulate?

Watch Me — Worked Example

"Points P and Q lie in plane M, so line PQ lies in plane M."
Postulate: if two points lie in a plane, then the **line containing them** lies in that plane.
Other postulates: two points $\rightarrow$ one line; three noncollinear points $\rightarrow$ one plane; two lines meet in one point; two planes meet in one line.

1. How many lines pass through points A and B?

exactly one

2. How many planes contain three noncollinear points J, K, L?

exactly one

3. Two different planes intersect. What is their intersection?

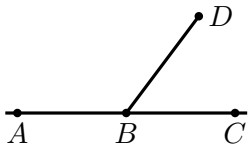
a line

4. Two different lines intersect. What is their intersection?

a point

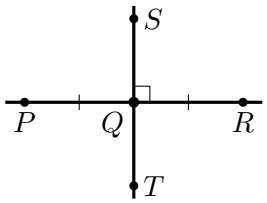
Type 2 What Can You Assume From a Diagram?

Watch Me — Worked Example



Yes: A, B, C are collinear; $\angle ABD$ and $\angle DBC$ are a linear pair (drawn).
No: $\angle DBC = 60°$, or $AB = BC$ — no marks say so.

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Use the diagram. Write **yes** or **no**.

5. P, Q, R are collinear. **yes**

6. $\overline{PR} \perp \overline{ST}$. **yes**

7. $PQ = QR$. **yes**

8. $SQ = QT$. **no**

Type 3 Sketch It

Watch Me — Worked Example

"Sketch two planes that intersect in line ℓ." Draw two flat sheets crossing each other and label the crossing line ℓ.
Tip: use tick marks and right-angle marks only for things that are given.

9. Sketch line m intersecting plane X at point A.

a line piercing the plane at A

10. Sketch $\overline{AB}$ with midpoint M (use tick marks).

AM and MB each with one tick

Type 4 Always, Sometimes or Never?

Watch Me — Worked Example

"Three points are coplanar." Through any three points there is at least one plane. **Answer: always.**
"Three points are collinear." Some are on one line, some are not. **Answer: sometimes.**

11. Two points are collinear.

always

12. Two lines intersect in two points.

never

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13. Four points are coplanar.

sometimes

14. Two planes intersect in a point only.

never

15. A line and a plane intersect in one point.

sometimes

16. HARD Through a line and a point not on it, how many planes pass?

exactly one

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